IT5351 - DATABASE MANAGEMENT SYSTEMS
MINI PROJECT

PRODUCT COMPANY

PROJECT REPORT

Submitted By

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INTRODUCTION

A Product Company Database Application is a software solution designed to efficiently manage and organize information related to a company's products. This type of application is essential for companies involved in manufacturing, retail, or distribution, as it helps streamline various processes, enhance decision-making, and improve overall operational efficiency. The database application serves as a centralized repository for storing and retrieving data about products, suppliers, customers, inventory, sales, and more.

Such an application is indispensable for companies seeking to enhance operational efficiency, make informed decisions, and stay competitive in today's dynamic business environment. It serves as a comprehensive solution to manage product-related data and support various aspects of the business lifecycle.

OBJECTIVES

Need for a Product Company Database Application:

1. Data Centralization:
   A product company deals with vast amounts of data, including product details, inventory levels, supplier information, and customer transactions. A centralized database application ensures that all this information is stored in one place, making it easier to manage and retrieve.

2. Efficient Inventory Management:
   Effective inventory management is crucial for optimizing stock levels, preventing overstock or stockouts, and reducing holding costs. A database application allows companies to track real-time inventory levels, manage stock movements, and implement automated reorder points.

3. Streamlined Order Processing:
   Handling customer orders, tracking order status, and managing shipments can be complex. A database application facilitates the automation of order processing, reducing errors, and ensuring timely fulfillment of customer orders.

4. Customer Relationship Management (CRM):
   Understanding customer preferences, purchase history, and interactions is vital for building strong customer relationships. A database application with CRM capabilities helps in managing customer data, analyzing buying patterns, and providing personalized services.

5. Supplier Management:
   For product companies, maintaining strong relationships with suppliers is crucial. The application can help manage supplier information, track deliveries, negotiate terms, and evaluate supplier performance.

6. Data Analysis and Reporting:
   The application provides tools for data analysis and reporting, allowing companies to gain insights into sales trends, product performance, and overall business analytics. This facilitates informed decision-making and strategy development.

7. Compliance and Security:
   Product companies often need to adhere to industry regulations and standards. A robust database application ensures data security, integrity, and compliance with regulatory requirements, protecting sensitive information.

8. Scalability:

As a company grows, its data requirements also increase. A scalable database application can adapt to the evolving needs of the business, accommodating more data, users, and functionalities.

BASIC IDEA BEHIND THE PROJECT

The idea behind a product company is centered on creating, marketing, and selling tangible or intangible products to consumers or businesses. Unlike service-based companies that primarily offer services, product companies focus on developing and delivering products that meet specific needs or solve particular problems.

TECHNOLOGIES USED

1. **HTML (HyperText Markup Language):**

- **Purpose:** HTML is the standard markup language used to create the structure of web pages and web applications. It defines the elements and their arrangement on a web page.
- **Key Features:**
  - Uses tags to structure content (headings, paragraphs, lists, links, etc.).
  - Provides a foundation for web development and is often used in conjunction with CSS and JavaScript.
  - HTML5, the latest version, introduced new elements and features for multimedia, graphics, and improved semantics.

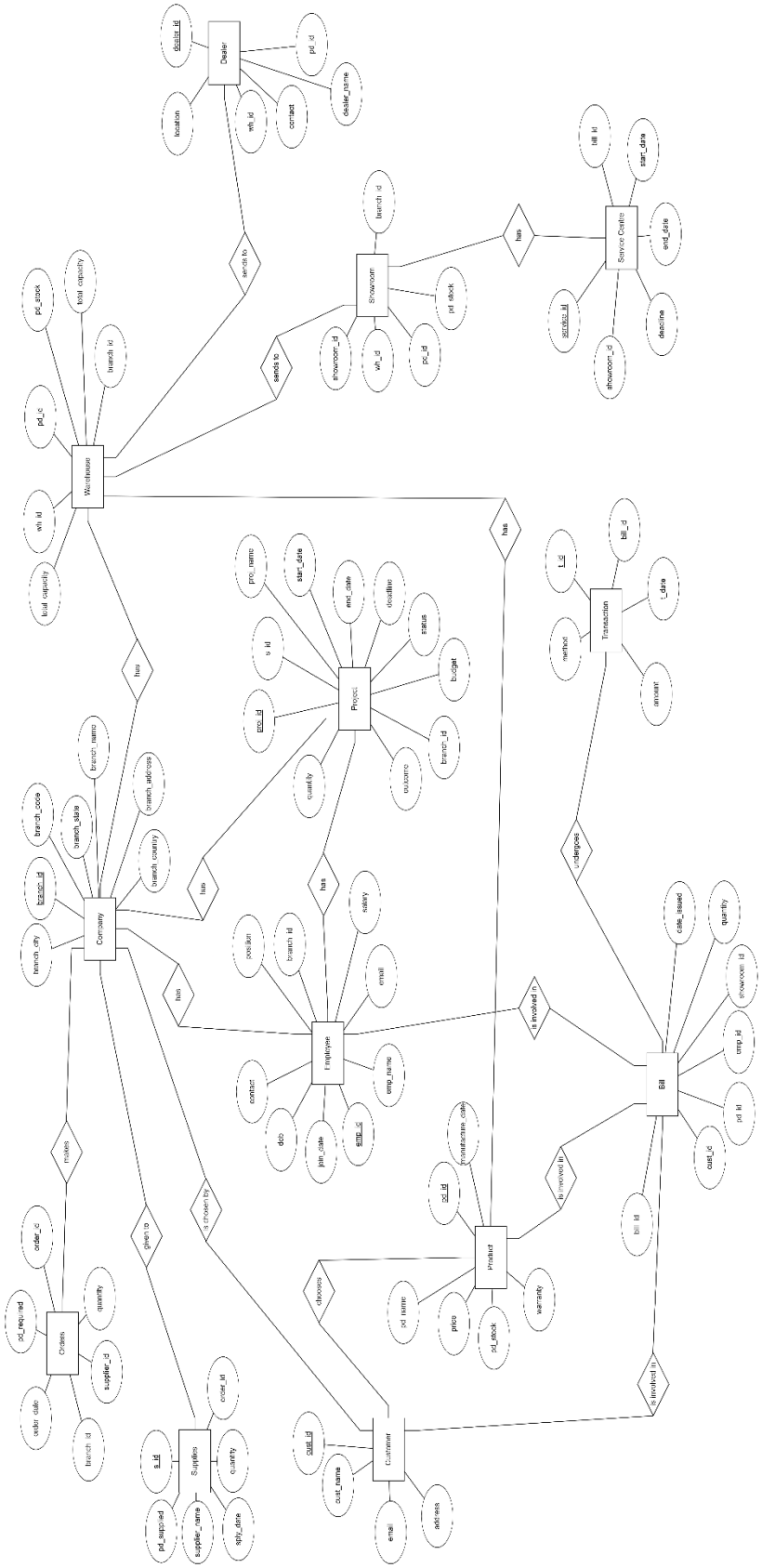
2. **CSS (Cascading Style Sheets):**
- **Purpose:** CSS is a style sheet language that controls the presentation and layout of HTML documents. It enhances the visual appeal and design of web pages.
- **Key Features:**
  - Selectors are used to target HTML elements and apply styles like colors, fonts, spacing, and positioning.
  - Supports responsive design, allowing developers to create layouts that adapt to different screen sizes and devices.

    - Can be applied internally within an HTML document, externally as a separate file, or inline directly within HTML tags.

3. **PHP (Hypertext Preprocessor):**
    - **Purpose:** PHP is a server-side scripting language designed for web development. It is embedded within HTML code and executed on the server, generating dynamic content for web pages.
    - **Key Features:**
        - Supports a wide range of databases, making it suitable for building database-driven web applications.
        - Integrates seamlessly with HTML, allowing developers to embed PHP code directly into HTML files.
        - Commonly used for tasks such as form processing, user authentication, and server-side scripting.

4. **MySQL:**
    - **Purpose:** MySQL is a relational database management system (RDBMS) that stores and manages structured data. It is often used in conjunction with PHP to create dynamic, database-driven web applications.
    - **Key Features:**
        - Provides a scalable and reliable platform for storing and retrieving data.
        - Supports SQL (Structured Query Language) for querying and manipulating data.

ER DIAGRAM


Dealer
dealer_id
location
wh_id
contact
dealer_name
pd_id
sends to
Warehouse
pd_stock
total capacity
branch id
pd_id
wh id
total capacity
sends to
Showroom
branch id
pd stock
showroom_id
wh_id
pc_id
has
Service Centre
bill id
start_date
end_date
describe
service_id
showroom_id
has
Company
branch_name
branch_address
branch_code
branch_state
branch_country
branch_id
branch_city
has
has
has
has
Project
pro_name
start_date
end_date
deadline
status
budget
s id
emp_id
quantity
branch_id
description
has
Employee
salary
position
branch id
email
contact
emp_name
dob
join_date
emp_id
makes
Orders
order_id
pd_required
quantity
supplier_id
order date
branch id
given to
is shown by
Supplies
order_id
s_id
quantity
pd_supplied
supplier_name
sply_date
chooses
Product
pd_id
manufacture_date
pd name
price
pd_stock
warranty
Customer
cust_id
cust_name
email
address
is involved in
is involved in
is involved in
Bill
cate_issued
quantity
showroom id
emp_id
pd id
cust_id
bill id
undergoes
Transaction
bill_id
t_date
method
amount
t_id

ORIGINAL SCHEMA

PRODUCT_COMPANY (company, supplies, project, employee, product, warehouse, dealer, showroom, customer, bill, transaction, service centre)

1) COMPANY (branch_id, branch_name, branch_address, branch_city, branch_code, branch_state, branch_country)
2) SUPPLIES (s_id, pd_supplied, quantity, sply_date, supplier_name, branch_id)
3) PROJECT (proj_id, proj_name, budget, start_date, deadline, end_date, s_id, project_outcome, quantity)
4) EMPLOYEE (emp_id, emp_name, contact, email, dob, position, salary, join_date, proj_id, branch_id, wh_id, showroom_id)
5) PRODUCT (pd_id, pd_name, manufacture_date, price, warranty, pd_stock, proj_id)
6) WAREHOUSE (wh_id, pd_id, pd_stock, sent_date, branch_id)
7) DEALER (dealer_id, dealer_name, location, contact, pd_id, wh_id)
8) SHOWROOM (showroom_id, wh_id, pd_id, pd_stock)
9) CUSTOMER (cust_id, cust_name, address, email)
10) BILL (bill_id, date_issued, cust_id, pd_id, emp_id, quantity, showroom_id)
11) TRANSACTION (t_id, bill_id, t_date, method, amount)
12) SERVICE CENTRE (service_id, showroom_id, bill_id, start_date, end_date,

SCHEMA AFTER NORMALISATION

```
CREATE TABLE company (
	branch_id INT PRIMARY KEY,
	b_name varchar(25) NOT NULL,
	b_address varchar(40) NOT NULL,
	b_city varchar(15) NOT NULL,
	b_code varchar(11) NOT NULL,
	b_state varchar(20) NOT NULL,
	b_country varchar(15) NOT NULL
);

CREATE TABLE spares (
	spare_id INT PRIMARY KEY AUTO_INCREMENT,
	part_name VARCHAR(50) NOT NULL
);

CREATE TABLE branch_spares (
	branch_id INT NOT NULL,
	spare_id INT NOT NULL,
	quantity INT NOT NULL
);


CREATE TABLE spares_usage (
	log_id INT PRIMARY KEY AUTO_INCREMENT,
	spare_id INT NOT NULL,
	quantity INT NOT NULL,
	log_date DATE NOT NULL,
	employee_id VARCHAR(8) NOT NULL,
	branch_id INT NOT NULL
);
```

```
CREATE TABLE suppliers (
	supplier_id INT PRIMARY KEY AUTO_INCREMENT,
	supplier_name VARCHAR(60) NOT NULL
);

CREATE TABLE spares_suppliers (
	spare_id INT NOT NULL,
	supplier_id INT NOT NULL
);

CREATE TABLE orders (
	order_id INT PRIMARY KEY AUTO_INCREMENT,
	order_date DATE NOT NULL,
	spare_id INT NOT NULL,
	quantity INT NOT NULL,
	supplier_id INT NOT NULL,
	branch_id INT NOT NULL,
	status VARCHAR(10) DEFAULT 'INCOMPLETE' NOT
NULL
);

CREATE TABLE supplies (
	supply_id INT PRIMARY KEY AUTO_INCREMENT,
	order_id INT NOT NULL,
	supply_date DATE NOT NULL,
	quantity INT NOT NULL
);
```

```
CREATE TABLE project (
	project_id INT PRIMARY KEY,
	project_name VARCHAR(50) NOT NULL,
	start_date DATE NOT NULL,
	deadline DATE NOT NULL,
	end_date DATE,
	allocated_budget INT NOT NULL,
	status VARCHAR(20) DEFAULT 'IN PROGRESS' NOT
NULL
);

CREATE TABLE project_branch (
	project_id INT NOT NULL,
	branch_id INT NOT NULL,
	start_date DATE NOT NULL,
	end_date DATE NOT NULL
);



CREATE TABLE project_spares (
	project_id INT NOT NULL,
	spare_id INT NOT NULL,
	quantity INT NOT NULL,
	use_date DATE NOT NULL
);
```

```
CREATE TABLE employee (
	employee_id VARCHAR(8) PRIMARY KEY,
	password VARCHAR(20) NOT NULL,
	fname VARCHAR(30) NOT NULL,
	lname VARCHAR(30) NOT NULL,
	position VARCHAR(15) NOT NULL DEFAULT 'Regular',
	email_id VARCHAR(50) NOT NULL,
	gender VARCHAR(1) NOT NULL,
	dob DATE NOT NULL,
	join_date DATE NOT NULL,
	salary INT NOT NULL,
	branch_id INT NOT NULL
);

CREATE TABLE employee_phone_no (
	employee_id INT NOT NULL,
	phone_no VARCHAR(15) NOT NULL
);

CREATE TABLE employee_branch (
	employee_id INT NOT NULL,
	branch_id VARCHAR(8) NOT NULL,
	join_date DATE NOT NULL
);


CREATE TABLE customer (
	phone_no VARCHAR(15) PRIMARY KEY,
	fname VARCHAR(30) NOT NULL,
	lname VARCHAR(30) NOT NULL,
	email_id VARCHAR(255) NOT NULL,
	points INT NOT NULL DEFAULT 0
);
```

```
CREATE TABLE product (
	product_id INT PRIMARY KEY,
	product_name VARCHAR(50) NOT NULL,
	specifications VARCHAR(255) NOT NULL,
	price INT NOT NULL,
	warranty VARCHAR(20) NOT NULL
);

CREATE TABLE project_product (
	project_id INT NOT NULL,
	product_id INT NOT NULL
);

CREATE TABLE manufactures (
	manufacture_id INT PRIMARY KEY AUTO_INCREMENT,
	project_id INT NOT NULL,
	product_id INT NOT NULL,
	manufacture_date INT NOT NULL,
	source_branch_id INT NOT NULL,
	quantity INT NOT NULL,
	status VARCHAR(8) NOT NULL DEFAULT 'NOT SENT'
);

CREATE TABLE transports (
	transport_id INT PRIMARY KEY AUTO_INCREMENT,
	manufacture_id INT NOT NULL,
	destination_branch_id INT NOT NULL,
	send_date DATE NOT NULL,
	receive_date DATE,
	status VARCHAR(12) NOT NULL DEFAULT 'NOT
RECEIVED'
);
```

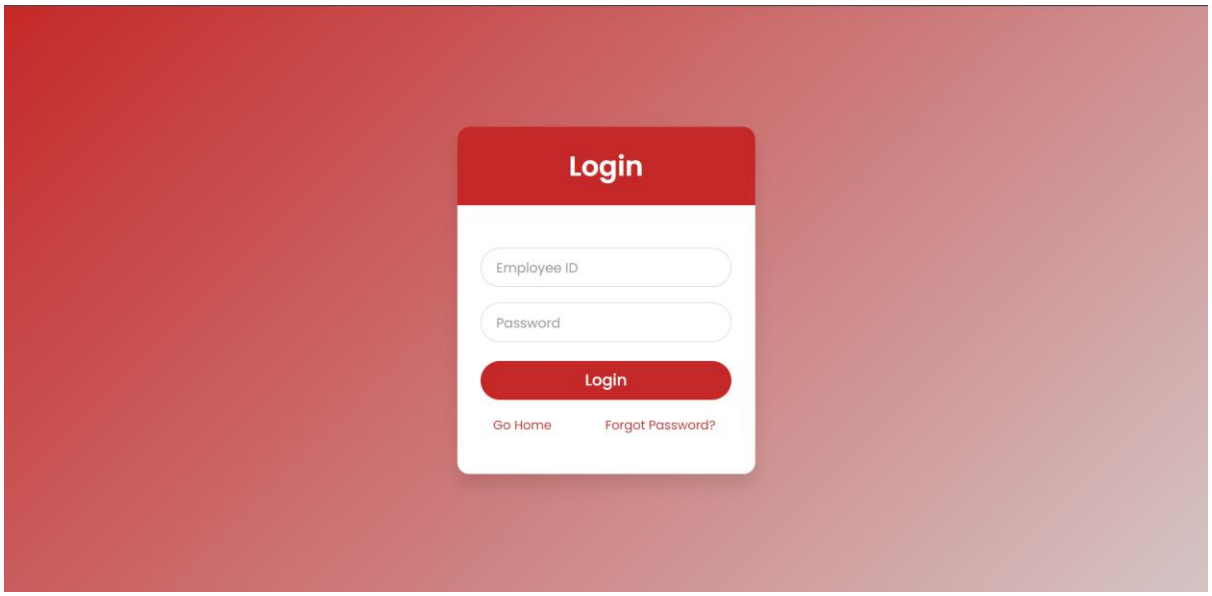
```
CREATE TABLE warehouse (
	branch_id INT NOT NULL,
	product_id INT NOT NULL,
	product_stock INT NOT NULL
);

CREATE TABLE showroom (
	branch_id INT NOT NULL,
	product_id INT NOT NULL,
	product_stock INT NOT NULL,
	manufacture_date DATE NOT NULL
);

CREATE TABLE dealer (
	dealer_id INT PRIMARY KEY AUTO_INCREMENT,
	dealer_name VARCHAR(20) NOT NULL,
	contact_no VARCHAR(15) NOT NULL,
	contact_mail VARCHAR(255) NOT NULL,
	d_address varchar(40) NOT NULL,
	d_city varchar(15) NOT NULL,
	d_code varchar(11) NOT NULL,
	d_state varchar(20) NOT NULL,
	d_country varchar(15) NOT NULL
);

CREATE TABLE warehouse_transports (
	warehouse_transport_id INT PRIMARY KEY
AUTO_INCREMENT,
	product_id INT NOT NULL,
	manufacture_date DATE NOT NULL,
	source_branch_id INT NOT NULL,
	destination_branch_id INT,
	destination_dealer_id INT,
	destination_type VARCHAR(8) NOT NULL,
	quantity INT NOT NULL,
	send_date DATE NOT NULL,
```

```
	receive_date DATE,
	status VARCHAR(12) NOT NULL DEFAULT 'NOT
RECEIVED'
);


CREATE TABLE dealer_product_branch (
	dealer_id INT NOT NULL,
	product_id INT NOT NULL,
	branch_id INT NOT NULL,
	quantity INT NOT NULL
);


CREATE TABLE bill (
	bill_id INT PRIMARY KEY AUTO_INCREMENT,
	customer_id VARCHAR(15) NOT NULL,
	employee_id INT NOT NULL,
	branch_id INT NOT NULL,
	date_issued DATE NOT NULL
);

CREATE TABLE bill_product (
	bill_id INT NOT NULL,
	product_id INT NOT NULL,
	quantity INT NOT NULL,
	manufacture_date DATE NOT NULL,
	status VARCHAR(6) NOT NULL DEFAULT 'UNPAID'
);
```

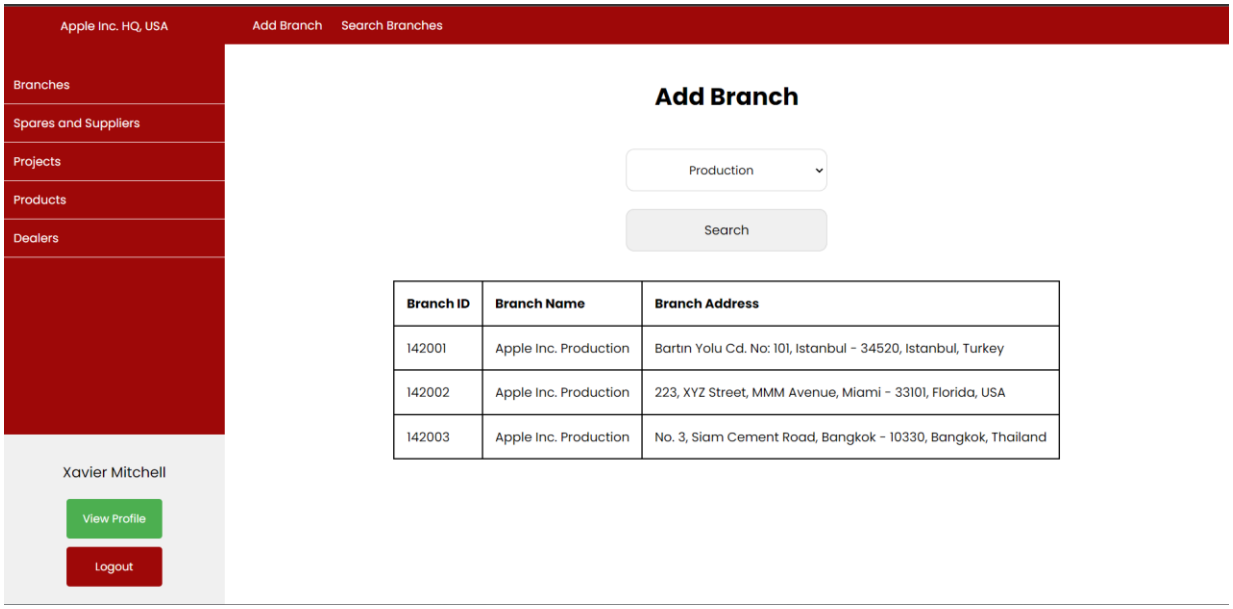
```
CREATE TABLE transaction (
	transaction_id INT PRIMARY KEY AUTO_INCREMENT,
	bill_id INT NOT NULL,
	transaction_date DATE,
	total_amount INT,
	payment_method VARCHAR(30) NOT NULL,
	status VARCHAR(14) NOT NULL DEFAULT 'NOT
SUCCESSFUL'
);

CREATE TABLE services (
	service_id INT PRIMARY KEY AUTO_INCREMENT,
	product_id INT NOT NULL,
	customer_id INT NOT NULL,
	employee_id VARCHAR(8) NOT NULL,
	problem_description VARCHAR(255) NOT NULL,
	branch_id INT NOT NULL,
	start_date DATE NOT NULL,
	deadline DATE NOT NULL,
	end_date DATE,
	status VARCHAR(10) NOT NULL 'INCOMPLETE'
);
```

WEBSITE

LOGIN PAGE



ADMIN PAGE



Apple Inc. HQ, USA

Create New Project | Search Projects | Add Project to Branch | Update Project

- Branches
- Spares and Suppliers
- Projects
- Products
- Dealers

Xavier Mitchell

View Profile

Logout

Search Project

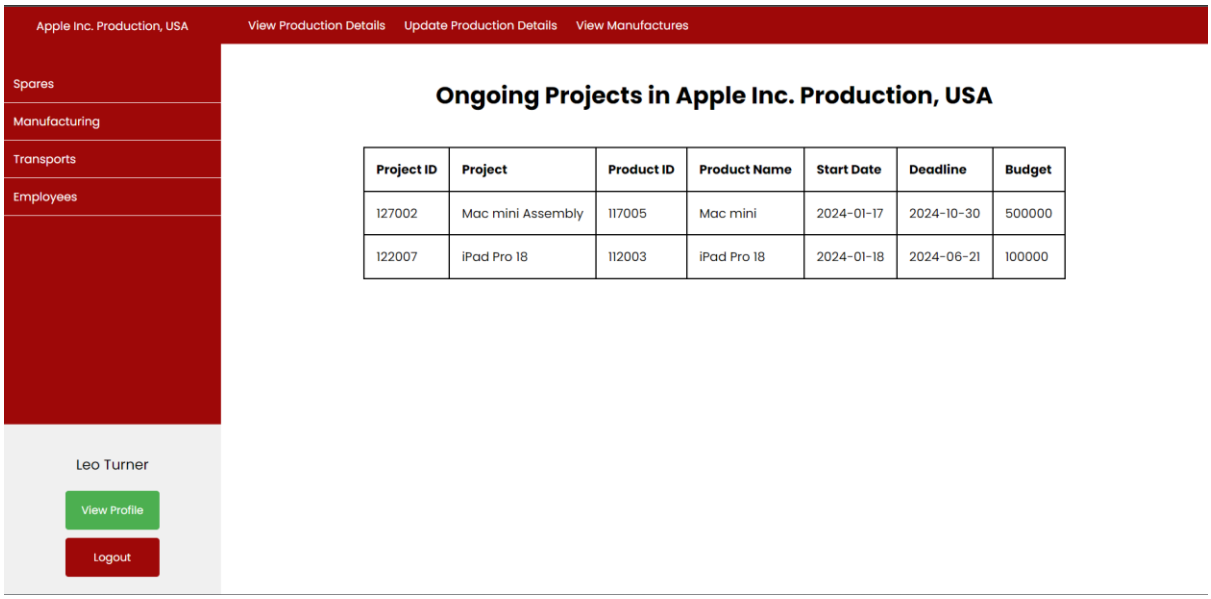
Status

Search Project

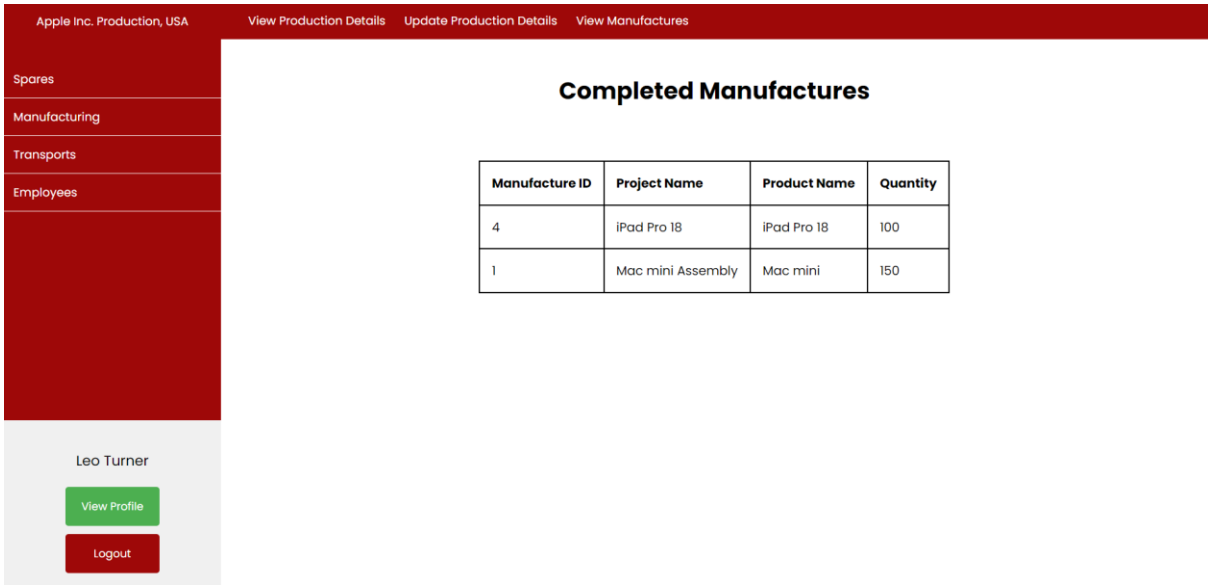
Search

Project ID	Project Name	Start Date	Deadline	Allocated Budget
121004	iPhone 16 Production	2023-12-30	2024-06-29	1000000

PRODUCTION PAGE

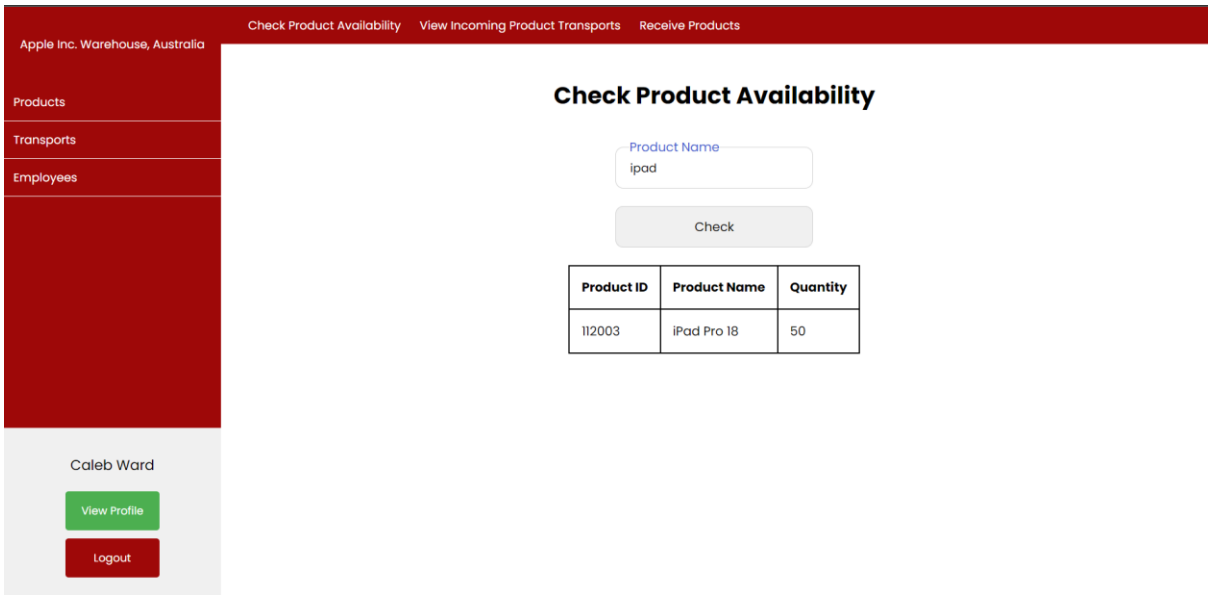


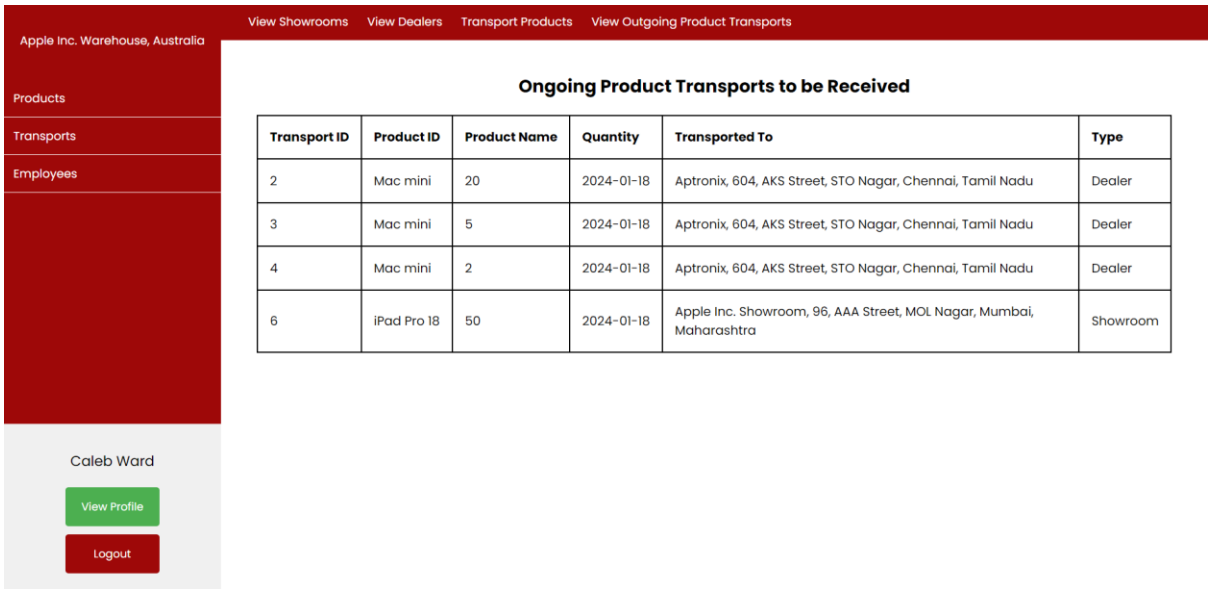
Project ID	Project	Product ID	Product Name	Start Date	Deadline	Budget
127002	Mac mini Assembly	117005	Mac mini	2024-01-17	2024-10-30	500000
122007	iPad Pro 18	112003	iPad Pro 18	2024-01-18	2024-06-21	100000



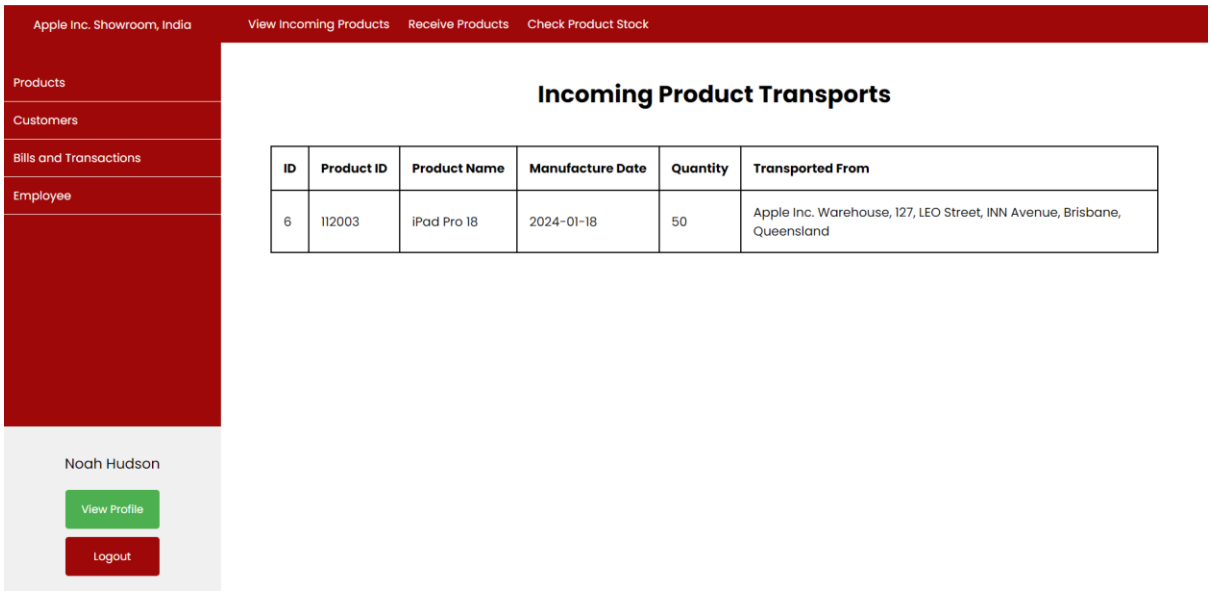
Manufacture ID	Project Name	Product Name	Quantity
4	iPad Pro 18	iPad Pro 18	100
1	Mac mini Assembly	Mac mini	150

WAREHOUSE PAGE

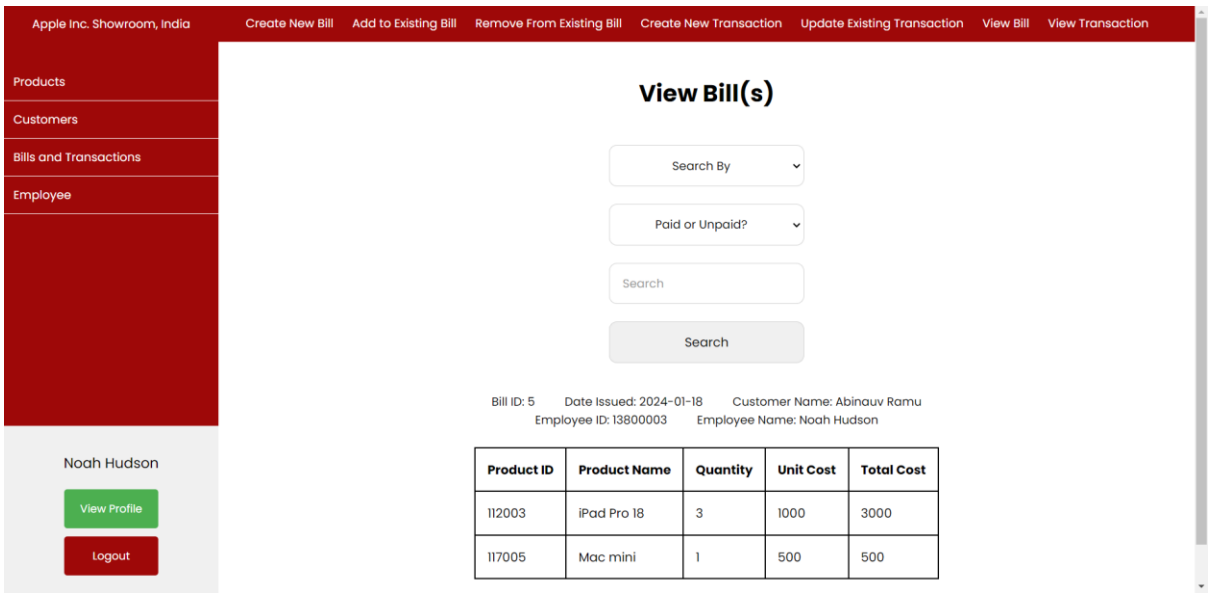




SHOWROOM PAGE

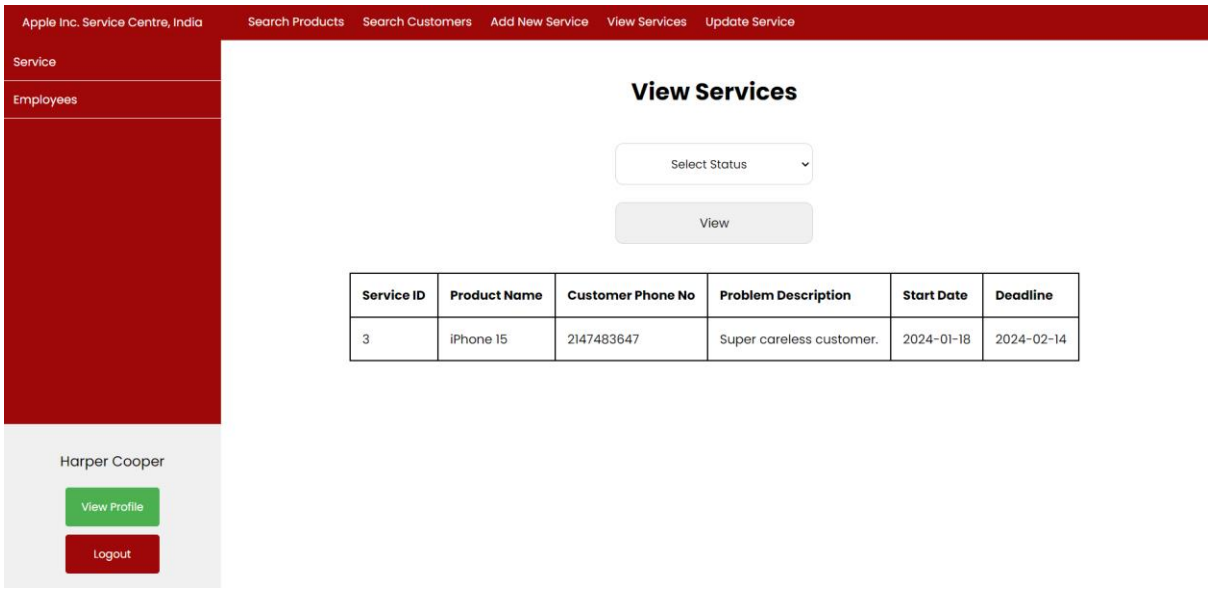


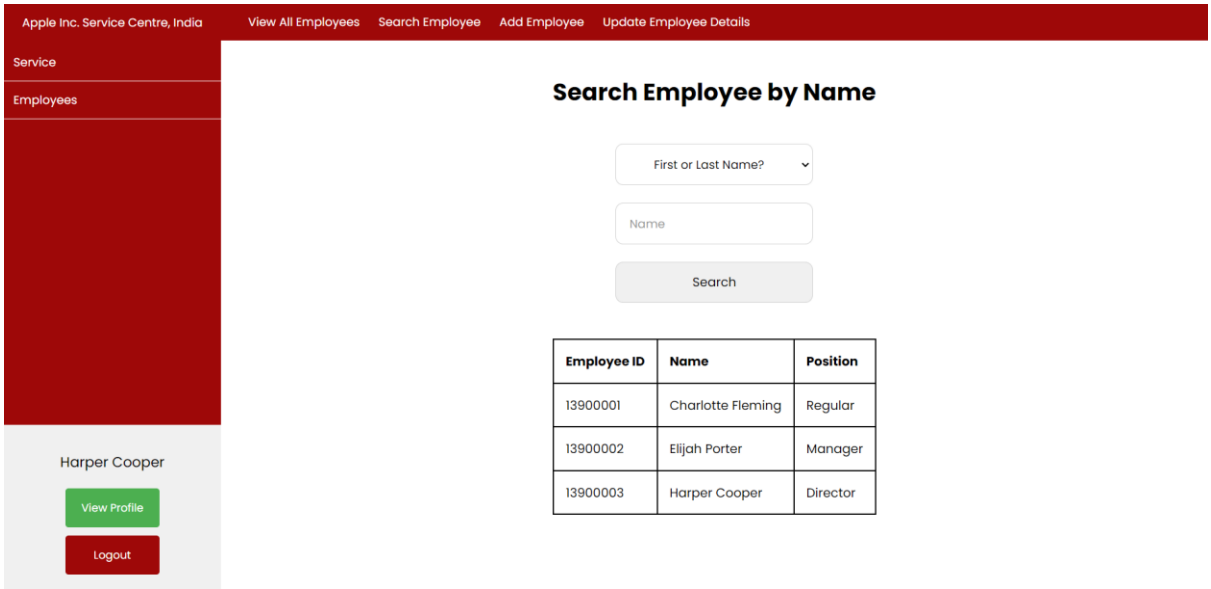
ID	Product ID	Product Name	Manufacture Date	Quantity	Transported From
6	112003	iPad Pro 18	2024-01-18	50	Apple Inc. Warehouse, 127, LEO Street, INN Avenue, Brisbane, Queensland



Product ID	Product Name	Quantity	Unit Cost	Total Cost
112003	iPad Pro 18	3	1000	3000
117005	Mac mini	1	500	500

SERVICES PAGE





CONCLUSION

In conclusion, the proposed Product Company Database Application is designed to address the specific needs of a company involved in manufacturing and distributing products across multiple branches. With a robust database schema, the application enables efficient management of branch information, product details, and inventory tracking. The integration of SQL queries allows for seamless retrieval and manipulation of data, enhancing the overall functionality and user experience.

The application's user-friendly interface facilitates easy data entry and retrieval, contributing to streamlined operations within the company. By leveraging a well-organized database, the application supports effective decision-making processes, inventory control, and reporting. The inclusion of primary and foreign keys ensures data integrity and establishes meaningful relationships between different entities.

Additionally, the application provides a foundation for future scalability and customization as the company's requirements evolve. It aligns with best practices in database design and management, promoting efficiency and reliability in handling business-critical information. Overall, the Product Company Database Application is poised to enhance operational efficiency, data accuracy, and decision-making capabilities, contributing to the overall success and growth of the company.